

Practical No. 01

Connecting Two Devices via a Network Wire on Cisco Packet Tracer

Easiest revision notes, written for someone starting with zero networking knowledge

What This Practical Is About

In this practical you take two computers and connect them directly with a cable inside Cisco Packet Tracer. Then you give each computer an address (called an IP address) and finally you test whether they can actually talk to each other. That is the whole practical in one line. Everything below just explains each part slowly.

Key Terms Explained First

Cisco Packet Tracer. A free simulator program made by Cisco. It lets you build a network on your screen using virtual computers, cables, switches and routers, without needing any real hardware. You practice networking safely inside this software before ever touching real equipment.

Network. A group of devices connected together so they can share information with each other, for example two computers sharing a file.

End device. Any device at the "end" of a network that actually sends or receives data, such as a PC, laptop or printer. It is different from devices like switches or routers, which just move data around without being the final sender or receiver.

IP address. A unique number given to a device so other devices can find it and send data to it. Think of it like a house address, if two devices are on the same street (same network), each one still needs its own house number so mail reaches the right place.

Subnet mask. A number that travels alongside the IP address. Its job is to tell the device which part of the IP address represents the network and which part represents the specific device. You do not need to calculate it in this practical, you just enter a standard value such as 255.255.255.0.

FastEthernet port. The physical socket on a computer or networking device where you plug a network cable in. In Packet Tracer it appears as a small port you click on before dragging a cable to it.

Cable / wire. The physical connector used to link two devices so data can flow between them. In Packet Tracer, cables are chosen from a menu and then drawn between two devices with the mouse.

Crossover cable. A special type of cable used to connect two similar devices directly to each other, for example PC to PC, or switch to switch. It is the cable type used in this practical because we are connecting two PCs to one another with no switch in between.

Straight through cable. A different type of cable used to connect two different kinds of devices, for example a PC to a switch. It is mentioned here just so you can tell it apart from a crossover cable, this practical itself uses crossover only.

PDU (Protocol Data Unit). A general name for a unit of data travelling across a network. In Packet Tracer, "Add Simple PDU" simply means sending a small test message from one device to another to check if the connection works, similar in spirit to a ping.

LAN (Local Area Network). A network that covers a small area, like a single lab, home or building. The two PCs connected in this practical form the smallest possible LAN, just two devices.

Setup, Step by Step

1. **Get the software.** Download and install Cisco Packet Tracer, then open it. You may be asked to log in with a Cisco Networking Academy account, or you can continue as a guest.
2. **Add two PCs.** At the bottom toolbar, click on the End Devices category, choose PC, then click on the empty workspace to place it. Do this twice so you have two separate PCs on the screen, usually labelled PC0 and PC1 by default.
3. **Pick the right cable.** Click on Connections at the bottom toolbar, then choose the crossover cable option, since we are linking two PCs of the same type directly.
4. **Connect the cable.** Click on the first PC and choose its FastEthernet0 port, then drag the cable across and click on the second PC, choosing its FastEthernet0 port as well. A green light usually appears on both ends once the link comes up.

5. **Assign an IP address to PC0.** Click on PC0, open its Desktop tab, then open IP Configuration. Enter an IPv4 address such as 192.168.1.1, and set the subnet mask to 255.255.255.0.
6. **Assign an IP address to PC1.** Repeat the same steps for the second computer, giving it a different address on the same network, such as 192.168.1.2.
7. **Test the connection.** From the top toolbar, click the Add Simple PDU icon, then click on PC0 followed by PC1. Check the Realtime box or the PDU list at the bottom, if it shows Successful, both computers are properly connected and correctly addressed.

If the test says Failed instead of Successful, the usual reasons are, the cable is not connected properly, the two PCs are on different networks because their IP addresses do not match the same subnet, or the IP address was typed incorrectly.

The Exercise Rule to Remember

When labelling and addressing your own devices for the exercise, the rule given is, name devices as RollNumber underscore DeviceName, and assign IP addresses using your own roll number in place of x, so PC1 becomes 192.168.x.1 and PC2 becomes 192.168.x.2. This is simply a way for the teacher to see that each student built their own version with their own roll number.

Viva Questions for This Practical

Q1. What is Cisco Packet Tracer?

A free network simulation tool from Cisco that lets you design, build and test computer networks virtually before working with real hardware.

Q2. What is an IP address?

A unique numerical address given to a device so it can be identified and reached by other devices on a network.

Q3. What is a subnet mask used for?

It works together with the IP address to show which part of the address is the network and which part is the individual device.

Q4. Why is a crossover cable used to connect two PCs directly?

Because crossover cables are meant for connecting two devices of the same type directly, while straight through cables are used between different types of devices, such as a PC and a switch.

Q5. What does a Successful result in the PDU test mean?

It means the test data reached the other device and came back properly, confirming that the cable connection and IP addressing are both correct.

Q6. What is a LAN?

A Local Area Network, meaning a network that connects devices within a small area such as a lab or a building.

Q7. What would happen if both PCs were given the same IP address?

This creates an IP conflict, the network cannot tell the two devices apart properly, and communication problems occur.

Q8. What is the difference between an end device and a networking device like a switch?

An end device such as a PC is the actual source or destination of data, while a networking device like a switch or router only helps move that data along, it is not the sender or receiver itself.

Important Things to Remember

- Two similar devices, such as PC to PC, are connected with a crossover cable. Two different devices, such as PC to switch, are connected with a straight through cable.
- Both devices must be on the same network range for communication to succeed, meaning their IP addresses should share the same pattern, only the last part should differ, for example 192.168.1.1 and 192.168.1.2.
- The subnet mask 255.255.255.0 is a very common default value used in small practice networks like this one.
- A green link light between two devices in Packet Tracer means the physical connection is active, but it does not guarantee the IP settings are correct, always confirm with a PDU test.
- This practical is the foundation for later practicals such as ARP, since it teaches the very basic building block, two devices, one link, and correct addressing.

Quick recall order: Setup software, place two PCs, connect with crossover cable, assign IP addresses, test with PDU.